Do Stopwords and Punctuation Matter? A Systematic Evaluation for Multi-Class Sentiment Classification of Movie Reviews

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1 Abstract

High frequency function words (stopwords) and punctuation are routinely removed in natural language processing, largely for the reason of reducing feature dimensionality and the assumption that these items contribute little discriminative information. These closed class items although carry stylistic, structural and sometimes affective information that may be relevant for tasks such as sentiment classification.
This paper analyses the impact of stopword and punctuation removal on sentiment classification performance using the IMDB62 movie review dataset containing 61,987 reviews, in which ratings from 1–10 are mapped to positive, negative and neutral sentiment categories. We evaluate two classifiers, a Linear Support Vector Classifier (LinearSVC) and a Complement Naive Bayes (ComplementNB) model, across five preprocessing scenarios: (1) all stopwords and punctuation retained, (2) only stopwords removed, (3) only punctuation removed, (4) both stopwords and punctuation removed, and (5) frequency discounting via a square root weighting scheme. To ensure statistically rigorous evaluation, each scenario is assessed using 10-fold stratified cross-validation, yielding ten independent performance samples per condition. Global differences across scenarios are tested with the Friedman test, and pairwise comparisons are conducted using Holm-corrected Wilcoxon signed-rank tests. Across both models, deletion-based preprocessing consistently and significantly degrades performance, while frequency discounting matches or exceeds the full-text baseline. These findings provide statistically validated evidence that high-frequency function words and punctuation encode meaningful sentiment information that should be retained rather than removed.

Keywords: Sentiment Analysis, Stopword Removal, Punctuation Analysis, Frequency Discounting, Support Vector Machines, ComplementNB, Text Preprocessing, Statistical Significance Testing, IMDB62 Movie Reviews

2 Introduction

Sentiment analysis is an important task in natural language processing (NLP). It is widely used in film reviews, social media monitoring and product feedback analysis. The main goal is to extract people's feelings and opinions from the text to help understand public opinion. As more and more people share comments online, the demand for algorithms to automatically detect emotions is increasing, and this field is developing rapidly.

In most sentiment classification systems, text preprocessing is a necessary first step. A very common practice is to remove stopwords and punctuation marks. There are usually two main reasons for this. Firstly, people assume that these high-frequency words and punctuation marks do not contain useful information to distinguish different emotions. Secondly, removing them can make the vocabulary smaller, so that the model will run faster. These ideas are widely accepted, and the removal of stopwords and punctuation has become a standard practice for many NLP processes. However, this common practice may deserve more careful examination. Although stopwords itself may not have a clear meaning, it is not safe to assume that these do not play a role in sentiment expression. If it is removed without caution, it may negatively impact the accuracy of sentiment classification. In addition, the results of previous research on this topic are also uneven. Some studies believe that it is beneficial, some studies point out

its disadvantages, and some studies find no significant effect. These inconsistencies show that the problem has not been really solved and requires more systematic research.

To make up for this gap, this study explores the impact of removing stopwords and punctuation on sentiment classification. Here we use the IMDB62 movie review dataset and conduct controlled experiments. We test two classifiers under five different preprocessing conditions to observe their performance changes. In addition to directly deleting words, this article also tries an alternative method called ”frequency discounting”, which adopts a square root weighting scheme to reduce the influence of stopwords and punctuation marks. This study aims to verify whether this method is a better text preprocessing scheme. In the rest of this article, we will first discuss the relevant research questions, and then review some of the previous works done in this field. Then we describe the data set and experimental settings, and discuss the results of the five different preprocessing methods. Finally, we summarise the research findings and put forward the future research direction.

3 Research Questions

The following research questions organise our empirical investigation. The central question driving this study is:

Main RQ: Does the removal of stopwords and punctuation affect sentiment classification performance in text analysis, and if yes, how?
To answer this, we break it down into five sub-questions, each corresponding directly to one experimental condition tested in this study.

RQ1: Does retaining all stopwords and punctuation produce the strongest sentiment classification performance compared to all other preprocessing conditions?
Hypothesis 1: The full-text condition is expected to produce the strongest classification performance across all five conditions. It preserves the complete linguistic signal including negation markers, affective function words and expressive punctuation.

RQ2: Does removing only stopwords improve, maintain, or degrade sentiment classification accuracy relative to the full-text baseline?
Hypothesis 2: Removing stopwords alone will degrade classification accuracy relative to the full-text baseline. Standard stopword lists include negation markers such as not, no and never, and intensity modifiers such as very and too, whose removal risks altering the apparent sentiment of a review.

RQ3: Does removing only punctuation improve, maintain, or degrade sentiment classification accuracy relative to the full-text baseline?
Hypothesis 3: Removing punctuation alone will produce a measurable degradation in classification accuracy. Affective punctuation marks such as exclamation points, question marks and ellipses encode evaluative intensity and emotional tone whose loss reduces the signal available to the classifier.

RQ4: Does removing both stopwords and punctuation together produce greater performance degradation than removing either one in isolation?
Hypothesis 4: The joint removal condition will yield the lowest classification accuracy of all five conditions. Removing both simultaneously eliminates two independent dimensions of affective signal, producing an additive degradation greater than either removal alone.

RQ5: Does frequency discounting via a square root weighting scheme outperform binary stopword removal, and does it approach or exceed full-text performance?
Hypothesis 5: Frequency discounting will outperform complete stopword removal and approach full-text performance. Reducing the weight of high-frequency items rather than deleting them retains affect-bearing function words as features, making it a more principled alternative to complete deletion.

Statistical significance tests (Friedman test and Holm-corrected Wilcoxon signed-rank tests) will be used to assess whether the above hypotheses are supported.

4 Literature Review

4.1 The Origins and Assumptions of Stopword Removal

Luhn (1958) proposed removing both the most and least frequent words from document collections as a part of its process for identifying significant words to improve retrieval efficiency. It is specifically mentioned that common,

frequently used words such as pronouns, propositions, etc. carry low discriminating value. These low value words are included in nearly every exclusion list in NLP toolkits. The idea that high frequency words carry low information value and add noise has been challenged by both theoretical and empirical work, yet the practice of removing them remains largely unchanged in many NLP pipelines.

4.2 Stopword and Punctuation Removal and Sentiment Classification

Symeonidis et al. (2018) evaluates sixteen text preprocessing techniques for sentiment analysis across two Twitter datasets and multiple classifiers. Removing punctuation consistently reduced accuracy, while reducing repeated punctuation improved performance, reflecting its affective role. Many stopwords were found to modify or convey sentiment, suggesting their removal should be reconsidered.
Saif et al. (2014) evaluates six stopword removal methods across six Twitter datasets. Pre-compiled stopword lists consistently harmed performance, suppressing negation markers and intensifiers while increasing data sparsity. Removing only singleton words reduced the feature space by 65% while maintaining accuracy, offering the best practical tradeoff. The key finding is that *discriminative value*, not raw frequency, should govern word removal, directly motivating frequency discounting over deletion.
Some studies find that stopword inclusion or exclusion has little effect on performance. DiPietro (2022) frames stopword selection as a classifier optimization problem, using iterative deletion to retain only words whose removal harms accuracy. The resulting list outperforms the NLTK standard stoplist and preserves high-frequency sentiment words such as *not*, *never*, and *very*. Though computationally expensive, this confirms that the optimal boundary is task-dependent, reinforcing the case for continuous frequency discounting over binary deletion.

4.3 Punctuation as an Information Channel

Darmon et al. (2018) provides a detailed analysis of punctuation sequences as a textual signal independent of lexical content. This paper demonstrates that punctuation patterns encode genre, syntactic rhythm and authorial habit, and that punctuation alone carries substantial discriminating information between text categories. For sentiment analysis, this is relevant because punctuation marks such as exclamation marks and ellipses carry affective load. Their reflexive removal, treated as equivalent to stopword removal in most NLP pipelines may constitute a measurable information loss, which we will investigate explicitly in our experimental design.

4.4 TF-IDF and Frequency-Sensitive Feature Weighting

Das et al. (2023) compares TF-IDF and N-Gram feature extraction across six classifiers on the IMDB and Amazon Alexa datasets, finding that TF-IDF consistently outperforms N-Gram and achieves a maximum accuracy of 93.81% with Random Forests. Das and Chakraborty (2018) extend this further by combining TF-IDF with a Next Word Negation (NWN) technique, tagging words immediately following negation cues as semantically inverted, and achieving 89.91% accuracy on IMDB, outperforming standard TF-IDF baselines. Wang and Manning (2012) show that simple uni-gram and bi-gram features which are weighted with log-count ratios and derived from Naive Bayes (the NB-SVM approach) achieve state-of-the-art sentiment classification. These findings show that how frequency information is used matters more than whether common words are deleted. Binary word presence often outperforms raw term frequency, suggesting diminishing returns from repeated occurrences and providing an empirical basis for sublinear frequency discounting. Furthermore, bi-grams such as ”not good” or ”barely watchable” aid sentiment classification, indicating that stopword removal destroys sentiment-bearing constructions. Together, these results motivate frequency discounting as a safer alternative to deletion.

4.5 The Square Root Rule as a Frequency Discounting Mechanism

The Square Root Law, as explained by Fielding and Liebeck (1975), was used to study voting systems. The law says that in larger groups where people vote randomly, the count of their votes doesn't increase linearly, it increases proportional to the square root of their size. In our project, we aim to apply the same idea on word frequencies. Frequency of common words like “the,” “is,” and “was” is very high and would dominate the model causing inaccuracies, while on the other hand if all such words are removed we could lose meaningful sentiment information. Thus, we apply the square root law to all word frequencies to reduce the influence of extremely common terms. This theoretical motivation is supported by Dogan and Uysal (2019). They analyze the effect of different term frequency factors across seven supervised term weighting schemes, finding that a square root based term frequency factor (SQRT_TF) outperforms both raw term frequency (TF) and logarithmic term frequency (LOG_TF). This motivates our investigation of whether frequency discounting is a more principled alternative to the outright deletion of stopwords and punctuation.

4.6 Sentiment Analysis: Broader Context

Strong evidence that deleting stopwords may cause problems partly comes from the research on negative words. Wiegand et al. (2010) points out that removing negations from the text may cause poor performance of sentiment classifiers. Councill et al. (2010) showed that removing punctuation affects the performance of strict rule-based classifiers which highly depend on words like ”not”, ”but also”, etc. Thus, it highlights that punctuation removal and stopword removal cannot be treated as two separate problems. Pang et al. (2002) demonstrated that both Naive Bayes and SVM achieve good performance on movie review sentiment classification, making them suitable for this task. Kumar et al. (2025) demonstrated the usefulness of SVC for text based sentiment analysis and the appropriateness of TF-IDF for down-weighting common, less significant phrases while up-weighting informative words. This study removes punctuation and stopwords as part of its pipeline, but it doesn't assess the effect of each step separately, so it is unclear if these removals improve or degrade classification accuracy. This gap motivates our hypothesis that stopword removal, punctuation removal and frequency discounting each have distinct effects on classifier performance.

4.7 Statistical Significance Testing Across Datasets

Demsar (2006) provides a statistical framework for comparing machine learning models across datasets, recommending repeated performance measurements via cross-validation rather than single estimates. This motivates our use of 10-fold cross-validation to obtain fold-wise samples across preprocessing scenarios. For comparisons involving more than two conditions, the author recommends the Friedman test as a non-parametric alternative to ANOVA, followed by post-hoc Holm-corrected Wilcoxon signed-rank tests to control family-wise error. Following this framework, we treat each fold as a paired observation, apply the Friedman test to detect overall differences, and use Holm-corrected Wilcoxon tests to identify which preprocessing strategies differ significantly.

4.8 Synthesis and Research Gap

The literature shows that stopwords and punctuation often carry useful sentiment information, and removing them can hurt accuracy by suppressing negation markers and affective function words. While TF-IDF reduces the weight of frequent terms, it cannot handle negation without additional processing. No prior study has jointly examined frequency discounting, punctuation removal, and stopword removal within a single framework. This work addresses that gap by individually evaluating each preprocessing choice and their combinations for sentiment classification.

5 Research Methods

5.1 Dataset Description

The dataset used in this research is a movie review dataset - IMDB62. It is publicly available on Hugging Face and was compiled originally by Seroussi et al. (2014). The dataset consists of 61,987 user generated movie reviews. Each review has six fields: reviewID, userID, itemID, rating, review title and the main review text. The rating field is a numerical value from 1-10, which is used as a label for review sentiment. For the purpose of this study, reviews with ratings 1-3 (5,933 reviews) are classified as negative, reviews with rating 4-6 (19,611 reviews) are classified as neutral and reviews with rating 7-10 (36,443 reviews) are classified as positive. These classifications are performed directly on the raw rating values without any preprocessing of the dataset.

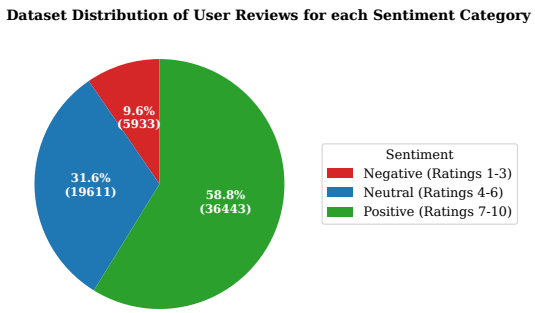

Figure 1: Distribution of User Reviews Across Sentiment Categories (Negative, Neutral, Positive)

The informal and natural style of the review text makes the IMDB62 dataset well suited for understanding the role of stopwords and punctuation in sentiment analysis. Stopwords, specifically words like ”no”, ”not” and ”but” that have negative notation can strongly affect sentiment analysis. Punctuation marks such as exclamation and question marks also contain information about emotions which is crucial for the task at hand. Since both these features appear consistently in the reviews, the dataset provides a strong foundation for evaluating whether removing stopwords and punctuation affects sentiment analysis.

Figure 2 provides an overview of the workflow used in this study, from preprocessing the IMDB62 dataset to model evaluation. Each component of the workflow is explained in detail in the following sections.

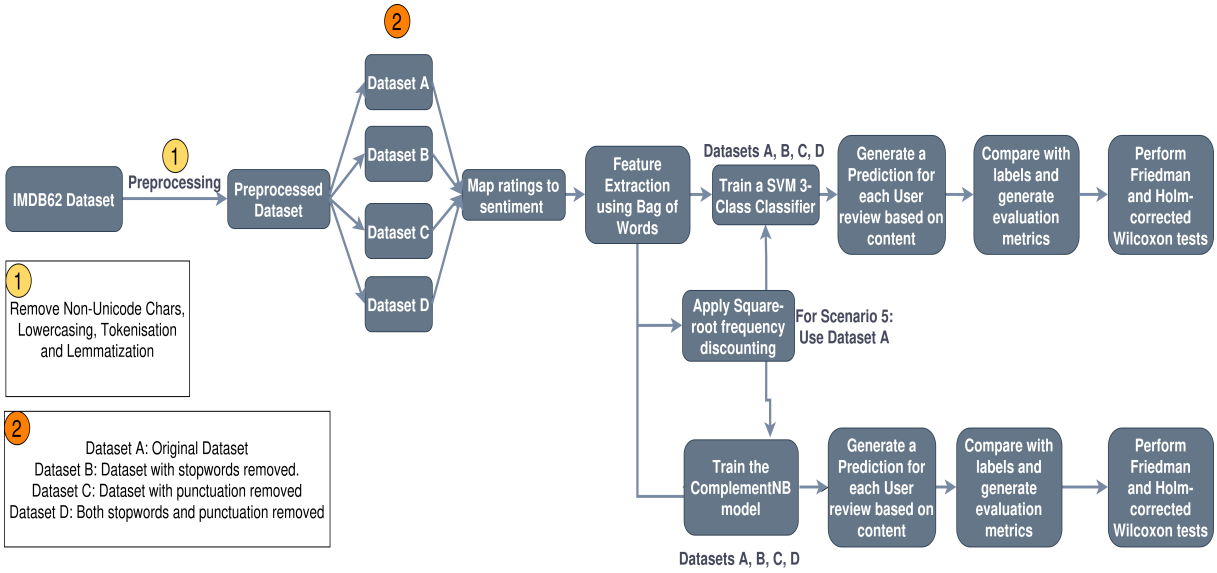

Figure 2: End-to-end Workflow for IMBD62 Sentiment Classification

5.2 Data Preprocessing

5.2.1 Dataset Preparation

To investigate the impact of stopwords and punctuation on sentiment analysis of movie reviews, three additional datasets were derived from the original IMDB62 dataset by applying the following transformations to the *title* and *content* columns:

1. **Stopwords Removed:** All stopwords were identified and removed using the standard stopword list provided by the NLTK library.
2. **Punctuation Removed:** All punctuation symbols, as defined by the `string.punctuation` list in Python, were identified and removed.
3. **Both Stopwords and Punctuation Removed:** Both stopwords and punctuation symbols were removed by combining the two transformations described above.

5.2.2 Text Preprocessing Pipeline

A consistent preprocessing pipeline was applied to the *title* and *content* columns across all datasets. The following pipeline was used for preprocessing:

1. **Removing Non-Unicode characters:** Non-Unicode characters (e.g., emojis) were identified using the regex pattern `r'[^\x00-\x7F]+'` and these were replaced with an empty string.
2. **Lowercasing:** All of the text was converted to lowercase using Python’s built-in `str.lower()` method.
3. **Stopword / Punctuation Removal:** Depending on the dataset variant, stopwords, punctuation, or both were removed.
4. **Tokenization:** The text was split into tokens using `word_tokenize` from the `nltk.tokenize` module.

5. **Lemmatization:** Each token was reduced to its base form using the `WordNet-Lemmatizer` from the `nltk.stem` module.

5.3 Model Description and Implementation

5.3.1 Overview

Using a classic machine learning pipeline, this paper performs a multi-class sentiment classification of movie reviews from the IMDB62 dataset. The method incorporates data preprocessing, Bag of Words feature extraction, frequency discounting and a linear Support Vector Machine (SVM) classifier. The reason why Support Vector Machines (SVMs) are preferred in text classification is because they can deal with high-dimensional sparse vectors very effectively and can effectively scale with large vocabularies. They are also robust to irrelevant features which are prevalent in the text. To assess whether the preprocessing effects generalise beyond a single classifier, we additionally evaluate the Naive Bayes Model (ComplementNB) across the same five scenarios using the identical 10-fold CV pipeline. ComplementNB is a variant of Multinomial Naive Bayes that estimates class probabilities from the *complement* of each class, which corrects for the class-imbalance bias that standard Multinomial NB exhibits. It requires no explicit `class_weight` parameter and has been shown to outperform standard Multinomial NB on imbalanced multi-class text classification tasks. Both models consume the same Bag-of-Words document–term matrix and, where applicable, the same square-root-discounted matrix for Scenario 5.

5.3.2 Data Loading and Sentiment Mapping

The process of modeling starts with the ingestion of the datasets in CSV format using a custom $load_data()$ procedure. The function has fault-tolerant parsing which can handle irregularities in the source files. The numerical ratings in every dataset, on the scale of 1-10 are encoded into three categories of sentiment:

- Negative (0): ratings 1–3
- Neutral (1): ratings 4–6
- Positive (2): ratings 7–10

This mapping helps to convert the ordinal rating variable into a categorical target variable which can be used for supervised classification. The review text itself is stored in the form of a string of a list of tokens.

5.3.3 Feature Extraction using Bag of Words

The Bag of Words model is used to generate textual features in Python with the help of the CountVectorizer() function. The vectoriser creates a sparse document-term matrix with each value being the frequency of a token in a document. A minimum document frequency of 5 is used to eliminate the rare terms which would not provide any meaningful contribution to the classification performance. This representation is the best for linear models and scales well even for large vocabularies.

5.3.4 Frequency Discounting

In order to test Hypothesis 5, we apply the square-root frequency discounting to the document-term matrix which is given by: $x \to \sqrt{x}$. To implement this scenario, we use the baseline dataset (with all stopwords and punctuation retained) and then apply the square-root weighting rule. This modification helps to reduce the impact of very common tokens without completely getting rid of them (Dogan and Uysal (2019)). This method is known to improve performance in text classification by reducing the dominance of high-frequency but low-information terms like stopwords.

5.3.5 Train-Test Split

To produce statistically reliable performance estimates, each experimental scenario is evaluated using **10-fold stratified cross-validation** rather than a single held-out split. In each fold, the data is partitioned into 90% training and 10% test subsets while preserving the original class distribution (Negative / Neutral / Positive) in every partition. The `CountVectorizer` is fitted exclusively on the training fold in each iteration, ensuring that no vocabulary information from the test fold leaks into feature construction. This procedure yields ten independent measurements of Accuracy, Macro-Precision, Macro-Recall, and Macro-F1 per scenario, which are subsequently used as paired observations for the statistical tests described in Section 5.3.6.

5.3.6 Statistical Significance Testing

Following the framework of Demsar (2006) for comparing classifiers across multiple conditions, we apply a two-stage non-parametric testing procedure to the ten fold-wise metric values obtained for each scenario.

Stage 1 - Friedman Test. The Friedman test is applied to the five vectors of fold-wise F1 scores (one vector per scenario). It ranks the five scenarios within each fold independently and tests whether the average ranks differ significantly across scenarios. The null hypothesis states that all five preprocessing conditions perform equivalently. A p-value below 0.05 leads us to reject this hypothesis and proceed to pairwise tests.

Stage 2 - Holm-corrected Wilcoxon Signed-Rank Tests. All $\binom{5}{2} = 10$ pairwise comparisons are carried out using the Wilcoxon signed-rank test, which is a paired non-parametric test that compares the fold-wise differences between two scenarios without assuming normality. To control the family-wise Type I error rate across the ten comparisons, we apply Holm correction, which first sorts the Wilcoxon p-values from smallest to largest and then tests them sequentially using a step-down procedure. A pairwise difference is declared significant only when the Holm-corrected p-value falls below 0.05.

5.3.7 Classification Models Used

a) **Linear Support Vector Machine (SVM):** Linear SVMs attempt to determine a hyperplane that optimally separates the classes i.e. maximize the margin between them (Cortes and Vapnik (2009)). For multi class classification, LinearSVC uses a one vs. rest strategy where it trains one classifier per class, each distinguishing that class from all others. Each classifier distinguishes one class from all others, and the final prediction corresponds to the class whose model yields the highest decision score (Fan et al. (2008)). For each class k, the SVM solves :

$$\min_{\mathbf{w}_k, b_k, \xi_i} \quad \frac{1}{2}\|\mathbf{w}_k\|^2 + C\sum_{i=1}^{N} \xi_i$$

subject to:

$$y_{i,k}(\mathbf{w}_k^\top \mathbf{x}_i + b_k) \geq 1 - \xi_i$$

where w: weight vector defining the hyperplane, b: bias term, ξ_i: slack variables allowing margin violations, C: regularisation parameter, $y_i \in \{-1, +1\}$: if sample i belongs to class k then 1 else -1 and $\mathbf{x}_i$: feature vector for sample i.
The final prediction is:

$$\hat{y} = \arg\max_k (\mathbf{w}_k^\top \mathbf{x} + b_k)$$

A $class_weight = 'balanced'$ parameter is passed during model definition as the IMDB62 dataset shows class imbalance. Here, Positive reviews are more common than negative and neutral ones. If the model is trained without adjustment it may over-predict the majority class and falsely achieve high accuracy while performing poorly on minority classes (He and Garcia (2009)). So class weighting automatically adjusts the weight of each class (w_k) according to:

$$w_k = \frac{N}{K \cdot n_k}$$

where N = total number of samples, K = number of classes and n_k = number of samples in class k.
So the minority classes are given higher weights while majority classes are given less weight. Hence misclassifying a minority-class sample causes a larger penalty, so the model is encouraged to learn decision boundaries that treat all classes fairly. This ensures consistent performance across all sentiment categories and improves the performance metrics like macro-averaged precision, recall and F1-score.

b) **Naive Bayes Model (ComplementNB):** Complement Naive Bayes (ComplementNB) is a variant of Multinomial Naive Bayes introduced by Rennie et al. (2003) that corrects for the class-imbalance bias inherent in standard Multinomial NB. The Multinomial model ComplementNB has been shown to perform well for text classification tasks (McCallum and Nigam, 1998). Whereas standard Multinomial NB estimates the likelihood of each word given a class k directly, ComplementNB instead estimates the likelihood of each word given the *complement* of uggingface.co/uggingface.co/class k, i.e. all documents *not* belonging to class k. For each class k and term t, the complement weight is computed as:

$$\hat{\theta}_{t,k} = \frac{\alpha + \sum_{i:\, y_i \neq k} x_{it}}{\alpha|V| + \sum_j \sum_{i:\, y_i \neq k} x_{ij}}$$

where x_{it} is the count of term t in document i, $|V|$ is the vocabulary size, and α is the Laplace smoothing parameter. The classification rule then assigns each document to the class whose complement model assigns it the *lowest* log-probability:

$$\hat{y} = \arg\min_k \sum_t x_t \log \hat{\theta}_{t,k}$$

This inversion of the decision rule means the model identifies the class for which the document is *least* explained by all other classes, which empirically produces better-calibrated estimates when class frequencies are unequal.

Since ComplementNB operates on raw or smoothed term counts rather than on a margin-based objective, it provides a qualitatively different inductive bias from LinearSVC. Evaluating the same five preprocessing scenarios across both classifiers therefore allows us to determine whether any observed effects of stopword or punctuation removal are artefacts of a particular model's sensitivity or are genuine properties of the preprocessing transformations themselves.

5.3.8 Model Evaluation

For each of the 10-fold cross validation splits, 90% data was used for training and remaining 10% was used for testing. Metrics like Accuracy, Macro-Precision, Macro-Recall and Macro F1-Score were calculated for each testing fold. These results were averaged with the total number of folds (10) and are mentioned in Tables 1 and 2.

5.3.9 Experimental Scenarios

Five experimental scenarios are executed to test each of the five hypotheses defined in section 3. These scenarios are: (1) All tokens are retained; (2) Only Stopwords are removed; (3) Only Punctuation is removed; (4) Both Stopwords and Punctuation are removed and (5) Frequency discounting applied (square root weighting). This design makes it possible to clearly evaluate how different preprocessing decisions can affect the performance of sentiment classification.

6 Result Discussion

The following results are based on 10-fold stratified cross validation. Tables 1 and 2 report the mean performance across the 10-folds recorded using the LinearSVC and ComplementNB models, with the standard deviation in parentheses. This allows for a statistical comparison across the five different preprocessing scenarios.

Scenario	Accuracy	Precision	Recall	F1-Score
S1: All retained	0.7158 (0.0078)	0.6113 (0.0129)	0.5981 (0.0123)	0.6037 (0.0114)
S2: Stopwords removed	0.6550 (0.0067)	0.5774 (0.0095)	0.5716 (0.0094)	0.5743 (0.0092)
S3: Punctuation removed	0.6575 (0.0069)	0.5833 (0.0088)	0.5776 (0.0099)	0.5801 (0.0090)
S4: Both removed	0.6535 (0.0062)	0.5758 (0.0078)	0.5697 (0.0083)	0.5725 (0.0079)
S5: Freq. Discounting	**0.7184 (0.0081)**	**0.6127 (0.0142)**	**0.5970 (0.0130)**	**0.6036 (0.0124)**

Table 1: LinearSVC: 10-fold CV performance (mean with std in parentheses)

Scenario	Accuracy	Precision	Recall	F1-Score
S1: All retained	0.6624 (0.0054)	0.5749 (0.0085)	0.5843 (0.0102)	0.5788 (0.0092)
S2: Stopwords removed	0.6678 (0.0049)	0.5826 (0.0089)	0.5858 (0.0099)	0.5835 (0.0094)
S3: Punctuation removed	0.6616 (0.0056)	0.5743 (0.0087)	0.5840 (0.0104)	0.5783 (0.0094)
S4: Both removed	0.6667 (0.0048)	0.5818 (0.0087)	0.5848 (0.0102)	0.5826 (0.0094)
S5: Freq. Discounting	**0.6689 (0.0051)**	**0.5852 (0.0071)**	**0.5870 (0.0087)**	**0.5847 (0.0080)**

Table 2: ComplementNB: 10-fold CV performance (mean with std in parentheses)

6.1 Statistical Test Results

LinearSVC: The Friedman test yields $\chi^2 = 33.44$, $p < 0.001$, providing strong evidence that the five preprocessing scenarios do not perform equivalently. Table 3 reports all ten pairwise Wilcoxon comparisons with Holm correction for the LinearSVC results.

Comparison	Raw p	Holm p	Significant?
S1 vs S2	0.0020	0.0195	Yes
S1 vs S3	0.0020	0.0195	Yes
S1 vs S4	0.0020	0.0195	Yes
S2 vs S5	0.0020	0.0195	Yes
S3 vs S5	0.0020	0.0195	Yes
S4 vs S5	0.0020	0.0195	Yes
S3 vs S4	0.0195	0.0781	No
S2 vs S3	0.0273	0.0820	No
S2 vs S4	0.2754	0.5508	No
S1 vs S5	**0.7695**	**0.7695**	**No**

Table 3: LinearSVC pairwise Wilcoxon signed-rank tests (Holm-corrected, $\alpha = 0.05$) on F1 score.

ComplementNB: The Friedman test yields $\chi^2 = 15.28$, $p = 0.0042$ ($p < 0.05$). This confirms that not all five preprocessing conditions are equivalent for ComplementNB, so we proceed to the post-hoc Wilcoxon pairwise comparisons reported in Table 4.

Comparison	Raw p	Holm p	Significant?
S2 vs S3	0.0020	0.0195	Yes
S1 vs S5	0.0039	0.0352	Yes
S3 vs S5	0.0039	0.0352	Yes
S3 vs S4	0.0098	0.0684	No
S1 vs S2	0.0137	0.0820	No
S1 vs S4	0.0371	0.1855	No
S4 vs S5	0.2324	0.9297	No
S2 vs S4	0.4922	1.0000	No
S2 vs S5	0.7695	1.0000	No
S1 vs S3	0.9219	1.0000	No

Table 4: ComplementNB pairwise Wilcoxon signed-rank tests (Holm-corrected, $\alpha = 0.05$) on F1 score.

6.2 Analysis

1. **Deletion-based preprocessing significantly degrades LinearSVC performance (Hypotheses 2 and 3 confirmed; Hypothesis 1 and 4 partially confirmed):** For the LinearSVC model, the Friedman test ($\chi^2 = 33.44$, $p < 0.001$) confirms that not all preprocessing conditions are equivalent. The Holm-corrected Wilcoxon tests show that Scenarios 2, 3 and 4 are each significantly worse than the full-text baseline Scenario 1 (all $p_{\text{Holm}} = 0.0195$). This provides a statistical backing for Hypothesis 2: stopword removal alone (Scenario 2) produces a large drop in mean F1 from 0.6037 (in full text baseline) to 0.5743, confirming that high-frequency function words such as negation markers carry discriminative signals. Punctuation removal (Scenario 3) also causes a significant decline, supporting Hypothesis 3. With respect to Hypothesis 4, the combined removal of stopwords and punctuation (Scenario 4) yields the lowest F1 score numerically (0.5725), consistent with the prediction that joint removal would be the most harmful condition. However, it is important to note that the three deletion scenarios (S2, S3, S4) are not significantly different from one another after Holm correction (all $p_{\text{Holm}} > 0.05$), indicating that their performance is statistically equivalent and that stopword removal is the dominant source of degradation. This means that although Hypothesis 4's expected effect is supported, the stronger claim that joint removal produces significantly greater harm than either individual removal lacks statistical support. Therefore Hypothesis 4 is only partially confirmed.
   The claim that full text retention (S1) produces the strongest performance across all five conditions is not supported. S1 achieves a mean F1 of 0.6037, which is statistically indistinguishable from Scenario 5 (F1 = 0.6036, p_{Holm} = 0.7695) but S1 is significantly better than all deletion scenarios. Hypothesis 1 is therefore partially confirmed for LinearSVC: full-text retention is among the best-performing conditions, outperforming all deletion strategies but not significantly better than S5 (Frequency Discounting). But for ComplementNB, H1 is not confirmed, as discussed in Finding 3 below.

2. **Frequency discounting is not statistically different from the full-text baseline for LinearSVC and significantly outperforms all deletion strategies (Hypothesis 5 confirmed):** Scenario 5 achieves a mean F1

of 0.6036 which is almost identical to Scenario 1 (0.6037) and the Wilcoxon test yields $p_{\text{Holm}} = 0.7695$, showing that the two conditions cannot be distinguished statistically. At the same time, Scenario 5 is significantly better than every deletion scenario (S2, S3, S4) (all $p_{\text{Holm}} = 0.0195$). This confirms Hypothesis 5: square-root frequency discounting retains affective function words as weighted features rather than removing them, thereby outperforming complete stopword removal and approaching full-text performance.

3. **ComplementNB shows greater robustness to preprocessing, but frequency discounting still provides a statistically significant advantage.** The ComplementNB model shows a notably different pattern. All five F1 scores lie within a narrow band (0.5783-0.5847), and most pairwise differences are not statistically significant after Holm correction. This indicates that the probabilistic structure of ComplementNB partially reduces the sensitivity to removing features. Nevertheless, Scenario 5 (Frequency Discounting) is the best performing condition (F1 Score: 0.5847) and is significantly better than the full text baseline Scenario 1 ($p_{\text{Holm}} = 0.0352$) and Scenario 3 ($p_{\text{Holm}} = 0.0352$). But this advantage does not extend uniformly to all conditions. S5 is not significantly better than Scenario 2 (pHolm = 1.000) or Scenario 4 (pHolm = 0.9297). So the benefit of frequency discounting for ComplementNB is selective rather than general. Unlike LinearSVC, where discounting and full-text retention showed equivalent performance, the ComplementNB model shows a measurable benefit from square-root weighting over the full text baseline, likely because normalising the dominant counts improves the multinomial likelihood estimates that Naive Bayes relies on. Hypothesis 1 is thus not confirmed for ComplementNB as full-text retention (S1) is significantly outperformed by frequency discounting (S5).

4. **Consistent finding across both models:** Despite different absolute performance levels (LinearSVC peaks at F1 = 0.6037 and ComplementNB at 0.5847), both models agree on the ranking that is,

$$Frequency\ Discounting \geq All\ Retained > Any\ Deletion\ Strategy$$

However, the magnitude of the gap between full retention or frequency discounting and the deletion strategies differs between the two classifiers. For LinearSVC, the differences between S1/S5 and the deletion scenarios are large and consistently statistically significant. For ComplementNB, the overall spread across all five conditions is narrow and most pairwise comparisons are non-significant, indicating that this model is considerably more robust to preprocessing choices. Despite this difference in sensitivity, the consistency of the ranking across both models strengthens the inference that the observed effects arise from the preprocessing choices themselves, rather than from any classifier specific bias.

7 Conclusion

Based on the experimental results from the IMDB62 dataset, it can be seen that directly deleting stopwords and punctuation produces an adverse effect on sentiment classification. By mapping ratings into three categories of sentiment labels and comparing them under different preprocessing conditions, this study shows that high frequency function words are not simply noise, but carry structural and tone information related to emotional expression.

Judging from the results, deleting stopwords or punctuation harms LinearSVC performance and shows mixed effects for ComplementNB. In contrast, the effect of frequency discounting is more stable. Instead of directly removing stopwords it reduces its impact while retaining the overall structure of the text. The experimental results show that this method is generally superior to the deletion scheme and shows strong feasibility in both the models.

The key finding of this study is that the sentiment analysis model relies not only on obvious emotional words, but also on the overall organisation of the text such as negative words, conjunctions, degree adverbs and punctuation marks with expressive functions. Although deleting these components can narrow the feature space, it may also lead to the loss of useful information. Therefore, for the task of sentiment analysis, instead of simply deleting high frequency words, it is better to reduce their influence using frequency discounting while retaining them. Future work could extend this analysis to datasets of different genres, languages and model frameworks to evaluate the generality of these preprocessing effects. The raw and preprocessed datasets for all scenarios are publicly available on Kaggle, and the source code is available in the GitHub Repository.

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Seroussi, Y., I. Zukerman, and F. Bohnert (2014). Authorship attribution with topic models. *Computational Linguistics 40*(2), 269–310.

Symeonidis, S., D. Effrosynidis, and A. Arampatzis (2018). A comparative evaluation of pre-processing techniques and their interactions for twitter sentiment analysis. *Expert Systems with Applications 110*, 298–310.

Wang, S. and C. Manning (2012). Baselines and bigrams: Simple, good sentiment and topic classification. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Jeju Island, Korea, pp. 90–94. Association for Computational Linguistics.

Wiegand, M., A. Balahur, B. Roth, D. Klakow, and A. Montoyo (2010). A survey on the role of negation in sentiment analysis. In *Proceedings of the Workshop on Negation and Speculation in Natural Language Processing*, pp. 60–68.

Individual Contributions - Group 2

1. **Drishya Dinesh (Chair):**

   - **Visualisations:** Designed all visualisations presented in Figures 1 and 2
   - **Literature Review:** Authored the following subsections:
     - *The Origins and Assumptions of Stopword Removal*
     - *Stopword and Punctuation Removal and Sentiment Classification* (first paragraph)
     - *Punctuation as an Information Channel*
     - *Sentiment Analysis: Broader Context* (last half)
     - *Statistical Significance Testing Across Datasets*
   - **Read the following papers to write the Literature Review:**
     - Symeonidis et al. (2018): A comparative evaluation of pre-processing techniques and their interactions for twitter sentiment analysis
     - Darmon et al. (2018): Pull out all the stops: Textual analysis via punctuation sequences.
     - Kumar et al. (2025): Sentiment Analysis of Movie Reviews Using TF-IDF
     - Luhn (1958): The Automatic Creation of Literature Abstracts
     - Demšar (2006): Statistical Comparisons of Classifiers over Multiple Data Sets
   - **Model Description and Implementation:** Authored the following subsections:
     - *Overview* (first paragraph)
     - *Data Loading and Sentiment Mapping*
     - *Feature Extraction using Bag of Words*
     - *Frequency Discounting*
     - *Linear Support Vector Machine (SVM)*
     - *Experimental Scenarios*
   - **Understanding and Explaining LinearSVC:** Reviewed and integrated theoretical and methodological insights from the following papers to write the model description for Linear SVC:
     - Cortes & Vapnik (2009): Support-vector networks
     - Fan et al. (2008): Liblinear: a library for large linear classification
     - He & Garcia (2009): Learning from imbalanced data
   - **Result Discussion and Analysis:** Wrote the entire Results Discussion and Analysis section, including interpretation of model behaviour and comparative evaluation.
   - **Proof-reading:** Collaboratively proof-read the final paper with the whole team.
   - In addition, I chaired all the group meetings, prepared meeting agendas and managed the team overall.

2. **Tanmay Ghosh (Verifier):**

   - **Initial Algorithm Implementation**: Implemented the initial LinearSVC sentiment classification pipeline covering all five experimental scenarios: all tokens retained, stopwords removed, punctuation removed, both removed, and square-root frequency discounting. This formed the foundational codebase for the group. Later, I implemented an updated Version of the pipeline (initially we thought of 2 - just Negative and Positive classes) introducing a three-class sentiment formulation (Negative / Neutral / Positive), incorporating class_weight='balanced' to address the inherent class imbalance in IMDB62, and reporting macro-averaged metrics across

all classes. Accompanied this with a detailed comparative analysis for the group for documentation of the impact of the binary-to-3-class transition in the paper.

- **Final Algorithm Implementation**: Updated the final project analysis pipeline which integrates with the shared cross-validation framework.
- **Shared Cross-Validation Library**: Designed and implemented cross-validation pipeline which is a model-agnostic, reusable 10-fold stratified cross-validation library providing functions with Holm step-down correction. This library was used by both the LinearSVC pipeline and the ComplementNB pipeline, and correctly avoids data leakage by fitting the CountVectorizer exclusively on the training fold in every iteration.
- **Results Generation**: Generated all CSV result files for the LinearSVC model: per-fold performance data, summary statistics, and statistical test outputs.
- **Statistical Methodology**: Proposed and designed the full statistical testing framework for the group : 10-fold stratified cross-validation to generate repeated measurements, the Friedman test as the global omnibus test, and Holm-corrected Wilcoxon signed-rank pairwise comparisons following Demsar (2006). This methodology directly addressed the peer review feedback requesting stronger statistical validation.
- **Research Methods - Paper Writing**: Worked on providing the Train-Test Split subsection (10-fold CV methodology, no-leakage guarantee) and the Statistical Significance Testing subsection (two-stage non-parametric testing procedure) in the final paper.
- **Results Discussion and Conclusion - Paper Writing**: Drafted the Results Discussion section and Conclusion section across two iterations, providing the analytical interpretation of all five experimental scenarios and both pipeline versions.
- **Post-Peer-Review Revisions**: Contributed in a comprehensive, section-by-section revision guide detailing all 9 updates required after peer review, including the rewritten Abstract, updated Keywords, the 10-fold CV methodology replacement, the new Statistical Significance Testing subsection, updated results tables (mean with std), statistical test result tables (Tables 3 and 4), the fully updated Analysis subsection, and the updated Conclusion.
- **Literature Review:** Read the following papers which helped in understanding of the project paper and the corresponding literature reviews:
    - DiPietro (2022): Quantitative stopword generation for sentiment analysis via recursive and iterative deletion. CoRR, vol. abs/2209.01519.
        - Contributed to the subsection "Stopword and Punctuation Removal and Sentiment Classification".
    - Saif, Fernandez, He, and Alani (2014): On stopwords, filtering and data sparsity for sentiment analysis of Twitter. Proc. LREC'14, pp. 810-817.
        - Contributed to the subsection "Stopword and Punctuation Removal and Sentiment Classification".
    - Wang and Manning (2012): Baselines and Bigrams: Simple, Good Sentiment and Topic Classification. Proc. ACL 2012 (Short Papers), pp. 90-94.
        - Contributed to the subsection "TF-IDF and Frequency-Sensitive Feature Weighting".
    - Rennie, Shih, Teevan, and Karger (2003): Tackling the poor assumptions of naive Bayes text classifiers. Proc. ICML-2003, pp. 616-623.

        - Provided the theoretical grounding for the ComplementNB model description in the Model Description and Implementation section.
    - Seroussi, Zukerman, and Bohnert (2014): Authorship attribution with topic models. Computational Linguistics, 40(2), 269-310.
        - Primary provenance reference for the IMDB62 dataset used across all five experimental scenarios; contributed to the Dataset Description section.
    - Kohavi (1995): A study of cross-validation and bootstrap for accuracy estimation and model selection. Proc. IJCAI-95, Vol. 2, pp. 1137-1143.
        - Foundational reference supporting the choice of 10-fold stratified CV in the. Train-Test Split subsection of Research Methods.
- **Verifier Role:** As Verifier, verified the weekly responsibilities of the Chair (meeting agendas), Recorder (meeting minutes), Monitor (article summaries), Ambassador, and Accountant (time records) throughout the project. Maintained oversight of research integrity across all group activities.
- **Peer Reviews:** Conducted and submitted detailed peer reviews for the mid-term papers of Groups allotted to me, providing structured feedback across all 21 review criteria. The peer reviews were distributed across the team members and I Reviewed peer reviews for our mid term paper allocated to me and noted down the required changes needed to be done for the final paper.
- **Repository and Submission Organisation:** Restructured the repository file layout. Managed the Repository and code branches. Created the final submission folder (TA-Submission/) with organised, clearly-labelled code files, CSV results, and a comprehensive README documenting the replicability archive. Added all Additional Materials to the submission archive, including weekly role-wise reports of all weeks.
- **Proof-reading:** Collaboratively proof-read the final paper with the whole team.

3. **Raghav Manish Gupta (Recorder):**

- **Research Questions and Hypotheses:** Authored the entire Research Questions section (Section 3), including framing the main research question, defining all five sub-questions (RQ1–RQ5), and formulating the corresponding hypotheses for each experimental scenario.
- **Literature Review:** Authored the following subsections:
    - *TF-IDF and Frequency-Sensitive Feature Weighting*
    - *The Square Root Rule as a Frequency Discounting Mechanism*
- **Read the following papers to write the Literature Review:**
    - Das and Chakraborty (2018): An Improved Text Sentiment Classification Model Using TF-IDF and Next Word Negation
    - Das et al. (2023): A Comparative Study on TF-IDF Feature Weighting Method and its Analysis using Unstructured Dataset
    - Dogan and Uysal (2019): On Term Frequency Factor in Supervised Term Weighting Schemes for Text Classification
- **Model Description and Implementation:** Authored the following subsections:
    - *Overview (second paragraph introducing ComplementNB)*

- ***Naive Bayes Model (ComplementNB):*** *Reviewed and integrated theoretical and methodological insights from Rennie et al. (2003): Tackling the poor assumptions of naive bayes text classifiers, to write the model description for ComplementNB.*

- **Result Discussion and Analysis:** Performed the statistical significance testing for the ComplementNB model, worked with Mukul on the Friedman test and Holm-corrected Wilcoxon signed-rank pairwise comparisons, and contributed the corresponding results to the Results Discussion section.
- **Abstract and Keywords:** Updated the abstract and keywords section to reflect the final experimental results, models used, and key findings of the study.
- **Meeting Minutes:** Recorded and maintained meeting minutes for all group meetings throughout the project.

4. **Parth Deshmukh (Accountant):**

Introduction: Added the second paragraph of the introduction section.
Literature Review Section: After the peer-reviews, it was noted that the literature review was spanning multiple pages. It was re-done by considering every member's content and made concise.
Literature Review Papers: Authored the following subsection:

a. Stopword and Punctuation Removal and Sentiment Classification

Read the following papers to write the Literature Review:

a. K. V. Ghag and K. Shah, "Comparative analysis of effect of stopwords removal on sentiment classification," 2015 International Conference on Computer, Communication and Control (IC4), Indore, India, 2015, pp. 1-6, doi: 10.1109/IC4.2015.7375527. keywords: {Support vector machines;Text mining;Computational modeling;Classification algorithms;Conferences;Computers;Feature extraction;Sentiment Classification;Pre-processing;Stopwords Removal;Term Weighting;Term Frequency;Term Presence},
b. Munková, D., Munk, M., Vozár, M. (2014). Influence of Stop-Words Removal on Sequence Patterns Identification within Comparable Corpora. In: Trajkovik, V., Anastas, M. (eds) ICT Innovations 2013. ICT Innovations 2013. Advances in Intelligent Systems and Computing, vol 231. Springer, Heidelberg.
c. Pradana, A., & Hayaty, M. (2019). The Effect of Stemming and Removal of Stopwords on the Accuracy of Sentiment Analysis on Indonesian-language Texts. Kinetik: Game Technology, Information System, Computer Network, Computing, Electronics, and Control, 4(4)

The following two papers were read to perform preprocessing on the dataset:

a. HaCohen-Kerner Y, Miller D, Yigal Y. The influence of preprocessing on text classification using a bag-of-words representation. PLoS One. 2020 May 1;15(5):e0232525. doi: 10.1371/journal.pone.0232525. PMID: 32357164; PMCID: PMC7194364.
b. Uysal, Alper Kürşat & Gunal, Serkan. (2014). The impact of preprocessing on text classification. Information Processing & Management. 50. 104 - 112. 10.1016/j.ipm.2013.08.006.

The following two papers were read to get familiar with the methods to be included in our paper and write the research methods section:

a. The Hitchhiker's Guide to Testing Statistical Significance in Natural Language Processing (Dror et al., ACL 2018).
b. Varma, Sudhir & Simon, Richard. (2006). Bias in Error Estimation When Using Cross-Validation for Model Selection." BMC Bioinformatics, 7(1), 91. BMC bioinformatics. 7. 91. 10.1186/1471-2105-7-91.

Data Preprocessing: Performed Data-Preprocessing and created the four datasets for all the downstream tasks. Also authored the same section in the paper. Suggested the mappings from ratings to reviews.

Research Methods: Wrote the research methods section for

a) Train-Test-Split.
b) Statistical significance testing (after peer reviews).
c) Model description.
d) Model evaluation.

Visualizations: Fixed both figures and especially figure 2 to follow academic conventions following Drishya's figures for reference.

Accountant: As I was the accountant, I collected times for every member's work and kept a record of everyone's time allocated for the module work for each week.

5. **Yilin Wen (Ambassador):**

Literature Review: Read the following papers and wrote the subsection on sentiment analysis, with particular attention to negation handling and the broader role of sentiment analysis in NLP:
Wiegand, M., Balahur, A., Roth, B., Klakow, D., & Montoyo, A. (2010): A survey on the role of negation in sentiment analysis. Proceedings of the Workshop on Negation and Speculation in Natural Language Processing, pp. 60-68. - Used this paper to explain why negation is a central linguistic challenge in sentiment analysis and why polarity cannot be interpreted reliably from isolated words alone.
Councill, I., McDonald, R., & Velikovich, L. (2010): What's great and what's not: learning to classify the scope of negation for improved sentiment analysis. Proceedings of the Workshop on Negation and Speculation in Natural Language Processing, pp. 51-59. - Used this paper to support the discussion of negation scope, showing that identifying where negation begins and ends can improve sentiment classification performance.
Al-Qablan, T. A., Mohd Noor, M. H., Al-Betar, M. A., & Khader, A. T. (2023): A survey on sentiment analysis and its applications. Neural Computing & Applications, 35(29), 21567-21601. - Used this paper to position our study within the wider sentiment analysis literature and to show the continued relevance of preprocessing decisions for sentiment classification tasks.
Conclusion Section: Contributed to the conclusion section of the paper, synthesising the sentiment analysis findings and linking them back to the paper's central argument that

seemingly low-information features such as stopwords, punctuation, and frequency patterns can still influence sentiment classification outcomes.
Ambassador Role: As Ambassador, communicated the progress of my assigned section to the group, coordinated with teammates to ensure that the sentiment analysis literature review and conclusion aligned with the overall argument of the paper, and helped relay updates and feedback clearly during the writing process.

6. **Mukul Ghare (Monitor):**

Dataset Research and Finalisation: Researched and identified the IMDB62 dataset as suitable for the study's experimental goals. Located the dataset on Hugging Face and proposed its use to the group for all downstream tasks.

Literature Review: Read the following papers and wrote the corresponding subsections:

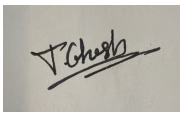
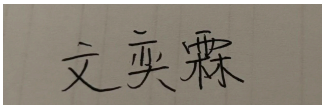
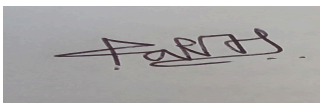
- Fielding, G. & Liebeck, H. (1975): Voting Structures and the Square Root Law. *British Journal of Political Science*, 5(2), 249–256. — Authored Section 4.5: *The Square Root Rule as a Frequency Discounting Mechanism*, providing the theoretical motivation for applying square-root weighting to word frequencies as an alternative to stopword deletion.
- McCallum, A. & Nigam, K. (1998): A Comparison of Event Models for Naive Bayes Text Classification. AAAI-98 Workshop on Learning for Text Categorization. — Provided theoretical grounding for the choice of the Multinomial event model underlying ComplementNB in the Model Description and Implementation section.
- Pang, B., Lee, L., & Vaithyanathan, S. (2002): Thumbs Up? Sentiment Classification Using Machine Learning Techniques. Proceedings of EMNLP 2002, pp. 79–86. — Supported the justification for using Naive Bayes alongside LinearSVC for movie review sentiment classification in the Model Description and Implementation section.

Model Description and Implementation: Implemented code for ComplementNB (Complement Naive Bayes) and worked with Raghav on doing classification pipeline across all five experimental scenarios, drawing on McCallum and Nigam (1998) for the Multinomial event model foundation and Pang et al. (2002) for its established suitability in movie review sentiment classification. This provided the second classifier used in the study, allowing the group to assess whether preprocessing effects generalised beyond LinearSVC.

Monitor Role: As Monitor, ensured all group members were contributing to the paper without overlap. Identified a duplication between Raghav's and Tanmay's literature review sections and flagged it to the group; Raghav subsequently revised his contribution to resolve the conflict.

Research Paper: Wrote literature review sections for my paper. Added keywords section after getting feedback from peer review. Collaboratively proof-read the final paper with the whole team.

Signatures

Team Member	Signature
Drishya Dinesh	
Tanmay Ghosh	
Mukul Ghare	
Yilin Wen	
Raghav Manish Gupta	
Parth Deshmukh